ASTDS Quiz 2

February 20, 2025

Name

Roll Number:

It's a 20 minutes close book/notes Quiz. Answer all questions.

1. Consider a multiple linear regression model
$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$,
where y, x_1 and x_2 represent the selling price (in lakhs INR), size (square feet)and age (years) of the apartments, respectively.

(i) Suppose the estimate of the marginal regression coefficient of x_1 on y is found to be positive and statistically significant and that of x_2 on y is statistically non-significant. However, the conditional estimates of the regression coefficients of x_1 and x_2 on y are found to be significant. Give an explanation of this phenomenon and state the expected signs of the regression coefficients.

- **Marginal Regression**: The marginal coefficient of x_1 (size) is positive and significant, indicating larger apartments have higher prices. The marginal coefficient of x_2 (age) is non-significant because age and size may be positively correlated thus, when the regression is run only on age, the age also proxies for size. The positive effect of size and negative effect of age cancel each other.
- **Conditional Regression**: When both x_1 and x_2 are included, the conditional coefficients of both size and age are significant. Size has a positive effect, while age has a negative effect due to depreciation.

Expected Signs

- β_1 (Size): Positive (larger the size of the apartments the price tends to increase on the average).
- β_2 (Age): Negative (older the apartments the price tends to decrease on the average).

(ii) Suppose a third variable x_3, the assessed price of the apartment, is added to the regression model. All three conditional regression coefficients are found to be non-significant. What could be an explanation of this result? How could you detect the problem statistically?

continued
When a third variable x_3 (assessed price) is added to the regression model, all three conditional regression coefficients (for size, age, and assessed price) may become non-significant due to **multicollinearity**. This occurs when independent variables are highly correlated, inflating the standard errors of the coefficients.

Multicollinearity can be detected using the **Variance Inflation Factor (VIF)**:

$$VIF = \frac{1}{1 - R_i^2},$$

where R_i^2 is the R-squared from regressing the i-th variable on the others.

$$VIF \geq 5 \quad indicates moderate multicollinearity.$$

$$VIF \geq 10 \quad indicates high multicollinearity.$$

(iii) Justify the following statement with proper reasons: Addition of a new variable in the multiple linear regression model always leads to an increase in the value of R-squared but may not be true for the adjusted R-squared.

Effect of Adding a New Variable on R^2 and Adjusted R^2

Adding a new variable to a multiple regression model **always increases** R^2, as it captures additional variance, even if the variable is irrelevant. However, this does not always improve model performance.

Unlike R^2, the **adjusted** R^2 accounts for model complexity. It penalizes unnecessary variables, so if the new variable does not significantly improve prediction, adjusted R^2 **may decrease**.

Thus, R^2 always increases, but adjusted R^2 only increases if the new variable adds real value.

continued

(iv) Suppose you have data on y, x_1 and x_2 for three cities viz., Mumbai, Delhi and Ahmedabad. Write down the linear regression model of y on x_1, x_2 using dummy variables for location. What kind of assumption this model makes regarding the effects of size and age on y? Does it always make sense?

((Points 5 + 5 + 5 + 5 = 20)

Model

To include the effect of location (Mumbai, Delhi, and Ahmedabad) in the regression model, we use dummy variables. Let:

- $D_1 = 1$ for Mumbai, 0 otherwise.
- $D_2 = 1$ for Delhi, 0 otherwise.
- Ahmedabad is the reference category (when $D_1 = 0$ and $D_2 = 0$).

The linear regression model of y on x_1, x_2, and location is:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 D_1 + \beta_4 D_2 + \epsilon,$$

where:

- β_0 is the intercept for Ahmedabad.
- β_1 and β_2 are the coefficients for x_1 (size) and x_2 (age), respectively.
- β_3 and β_4 are the effects of Mumbai and Delhi relative to Ahmedabad.
- ϵ is the error term.

Assumptions

This model assumes:

- The effects of x_1 (size) and x_2 (age) on y are **the same across all cities**. This is because the coefficients β_1 and β_2 are constant and do not vary by location.
- The only difference between cities is captured by the intercepts (β_3 and β_4).

This assumption may not always make sense. For example:

- If the effect of size (x_1) or age (x_2) on y differs by city, the model will be misspecified. In such cases, interaction terms (e.g., $x_1 \cdot D_1$, $x_2 \cdot D_2$) should be included to allow the effects of x_1 and x_2 to vary by location.